

PROJECT REPORT

ON

“EduGenie: Google Gemini Powered Learning Assistant”

A Project Report Submitted in Partial Fulfillment of the Requirements for the Degree of

BACHELOR OF COMPUTER APPLICATIONS (BCA)

Submitted By

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CERTIFICATE

This is to certify that the project entitled “EduGenie: Google Gemini Powered Learning Assistant” is a bonafide work carried out by the above-mentioned students in partial fulfillment of the requirements for the award of the Bachelor of Computer Applications (BCA) degree.

The project has been completed under my guidance and supervision during the academic year 2026- 2027.

Project Guide Signature
KAVIYA S

Head of the Department Signature
_____S_SIVARANJANI

DECLARATION

We hereby declare that the project report entitled “EduGenie: Google Gemini Powered Learning Assistant” is our original work carried out as part of the Bachelor of Computer Applications programme.

We further declare that this project has not been submitted previously, either fully or partially, to any other institution or university for the award of any degree or qualification.

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We would like to express our sincere gratitude to everyone who supported us in completing the project entitled “EduGenie: Google Gemini Powered Learning Assistant.”

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ABSTRACT

EduGenie is an AI-powered learning assistant designed to support students in understanding academic concepts and preparing for their studies. The system uses Google Gemini models to provide conversational explanations, summaries, examples and learning assistance based on student questions.

The application allows students to enter questions in natural language and receive clear, structured responses. It can provide topic explanations, short notes, question-and-answer support and revision assistance.

EduGenie aims to make learning more interactive and accessible through a simple AI-assisted study environment. It can later support quizzes, flashcards, progress tracking and multilingual learning.

TABLE OF CONTENTS

1. Introduction -	7
2. Existing System -	8
3. Proposed System -	9
4. System Requirements -	10
5. System Analysis -	11
6. System Design -	12
7. Modules -	13
8. Technologies Used -	14
9. Database Design -	15
10. Implementation -	16
11. Testing -	17
12. Results and Screenshots -	18
13. Advantages, Limitations&Future Scope -	19
14. Conclusion and References -	20

CHAPTER 1 – INTRODUCTION

Background

Students often use textbooks, websites, videos and other resources to understand academic topics. Finding the right explanation and revising large amounts of information can take considerable time.

Problem Statement

Students may face difficulty understanding complex concepts, obtaining quick explanations and organizing study materials for revision. A conversational AI assistant can provide immediate learning support.

Objectives

- To develop an AI-powered learning assistant for students.
- To provide simple explanations for academic questions.
- To use Google Gemini models for conversational learning support.
- To generate summaries, examples and revision notes.
- To provide an interactive study environment.

Scope

The system can support question answering, concept explanation, summarization, revision assistance, quizzes and personalized learning.

CHAPTER 2 – EXISTING SYSTEM

Existing Method

Students commonly depend on textbooks, search engines, videos, notes and teachers. They may need to search through multiple resources to find an explanation suitable for their level.

Limitations

- More time spent searching.
- Explanations may not match the student's level.
- Information is spread across multiple resources.
- Immediate question-and-answer support may not be available.
- Manual preparation of revision notes is time-consuming.

Need for AI Assistance

Generative AI can provide conversational explanations based on prompts and can adapt responses to formats such as short notes, examples and step-by-step explanations.

CHAPTER 3 – PROPOSED SYSTEM

Proposed System

EduGenie provides a single platform where students can ask academic questions and receive AI-assisted responses using Google Gemini models.

Main Features

- Student registration and login.
- Natural-language question input.
- AI-powered concept explanation.
- Summary and short-note generation.
- Examples and step-by-step explanations.
- Revision assistance.
- Chat history and saved learning content.

Workflow

Student Question → Input Processing → Gemini Model → AI Response → Display Explanation → Save / Review

CHAPTER 4 – SYSTEM REQUIREMENTS

Hardware Requirements

- Processor: Dual Core or above
- RAM: Minimum 4 GB
- Storage: Minimum 10 GB free
- Display: 1366 × 768 or higher
- Internet connection for Gemini API.

Software Requirements

- Windows/Linux/macOS
- HTML, CSS, JavaScript or React
- Python Flask/Django or Node.js
- MySQL/PostgreSQL/SQLite
- Google Gemini API
- Chrome/Edge/Firefox.

User Requirements

Basic knowledge of web applications and clear academic questions.

CHAPTER 5 - SYSTEM ANALYSIS

Functional Requirements

- User registration and login.
- Accept academic questions.
- Send suitable prompts to Gemini.
- Display AI-generated learning responses.
- Provide summaries and explanations.
- Save or review learning content.

Non-Functional Requirements

- Usability
- Performance
- Security
- Reliability
- Maintainability.

Feasibility

The project is technically feasible because web technologies, databases and cloud-based generative AI APIs can be integrated to develop an interactive learning assistant.

CHAPTER 6 - SYSTEM DESIGN

System Architecture

[STUDENT] → [WEB INTERFACE] → [BACKEND SERVER] → [GEMINI API / AI PROCESSING] → [LEARNING RESPONSE] → [DISPLAY / SAVE / REVIEW]

Use Case

Student logs in, enters a question, system validates it, backend sends a request to Gemini, response is generated and displayed, and student can save or continue the conversation.

Data Flow

The question moves from frontend to backend, the backend communicates with Gemini, and the generated response is returned to the interface.

CHAPTER 7 – MODULES

User Module

Manages student registration, login and profile information.

Question Input Module

Provides an interface for academic questions.

AI Learning Module

Uses Gemini models to generate explanations, summaries and examples.

Study Notes Module

Generates and saves revision notes.

Chat History Module

Stores previous learning conversations.

Quiz / Practice Module

Can generate basic practice questions.

Feedback Module

Allows users to provide feedback.

CHAPTER 8 – TECHNOLOGIES USED

Frontend

- HTML, CSS and JavaScript can build the interface
- React can be used for a component-based application.

Backend

Python Flask/Django or Node.js can manage authentication, API requests and application logic.

Google Gemini

Gemini models can support question answering, explanations, summarization, examples and educational content generation.

Database

SQLite, MySQL or PostgreSQL can store users, chats, notes and learning history.

Security

API credentials should be stored securely on the server and user information should be protected.

CHAPTER 9 - DATABASE DESIGN

Tables

Users: user_id, name, email. Chats: chat_id, user_id, question, response, created_at. Notes: note_id, user_id, content.

Relationship

One user can have multiple chat records and multiple saved notes. Users therefore has one-to-many relationships with Chats and Notes.

CHAPTER 10 – IMPLEMENTATION

Steps

- Create frontend and backend
- create database tables
- develop login
- create chat interface
- configure secure Gemini API
- design prompts
- process responses
- implement save/history
- test the application.

Sample Workflow

A student asks “Explain polymorphism in simple words.” EduGenie sends a suitable request to Gemini and displays a clear explanation. The student can ask for an example, short notes or revision questions.

Security

API keys should remain confidential. Authentication, validation and secure database access should be implemented.

CHAPTER 11 – TESTING

Testing Objective

Testing ensures the learning assistant performs expected functions and handles invalid input and API errors.

Test Cases

- TC01 Valid login → Dashboard
- TC02 Invalid login → Error
- TC03 Valid question → AI response
- TC04 Empty question → Validation
- TC05 Summary request → Summary
- TC06 Save note → Note stored
- TC07 Chat history → Previous chat
- TC08 API failure → Error message.

Testing Types

Unit testing, integration testing, system testing and user acceptance testing.

CHAPTER 12 – RESULTS AND SCREENSHOTS

Result

EduGenie provides an interactive learning environment where students can ask questions and receive AI-assisted explanations. It can support summaries, examples, revision notes and saved content.

Expected Screens

- Login / Registration
- Dashboard
- Question and Chat
- AI Response
- Summary / Notes
- Quiz / Practice
- Chat History
- Saved Notes.

Screenshot Placeholders

[DASHBOARD SCREENSHOT]

[AI CHAT SCREENSHOT]

[SAVED NOTES SCREENSHOT]

CHAPTER 13 – ADVANTAGES, LIMITATIONS&FUTURE SCOPE

Advantages

- Quick learning assistance.
- Simple conversational interface.
- Different levels of explanation.
- Summaries and revision material.
- Can be extended with quizzes and personalized learning.

Limitations

- AI responses may occasionally be inaccurate.
- Internet may be required.
- API usage may have limits or costs.
- Important information should be verified.

Future Enhancements

- Personalized learning paths.
- Voice-based assistant.
- Multilingual support.
- Automatic quizzes and flashcards.
- Progress tracking.
- Educational platform integration.

CHAPTER 14 – CONCLUSION AND REFERENCES

Conclusion

EduGenie demonstrates how generative AI can provide interactive learning assistance. By combining a simple interface with Google Gemini models, the system can answer questions, explain concepts, generate summaries and support revision.

Future

The project can be improved with personalized learning, quizzes, flashcards, voice interaction, multilingual support and progress tracking.

References

- Google AI for Developers – Gemini API documentation
- Google Gemini developer resources
- MDN Web Docs
- React Documentation
- Python Documentation
- MySQL Documentation.